

RAJALAKSHMI INSTITUTE OF TECHNOLOGY

Department of Mechanical Engineering

ME23312 ENGINEERING THERMODYNAMICS

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3	1	0	4

COURSE OBJECTIVES:

- To enable the students to understand the knowledge on the basics and application of zeroth and first law of thermodynamics.
- To equip the students to gain knowledge on the second law of thermodynamics, availability and applications of it.
- To impart knowledge to understand and apply the properties of pure substances and process.
- To facilitate the students in acquiring the knowledge on the macroscopic properties of ideal and real gases.
- To foster the ability among the students to analyze the properties of gas mixtures and Psychrometric processes.

UNIT I BASICS, ZEROTH AND FIRST LAW

9+3

Review of Basics – Thermodynamic systems, Properties and processes, Thermodynamic Equilibrium - Displacement work - P-V diagram. Thermal equilibrium - Zeroth law – Concept of temperature and Temperature Scales. First law – application to closed and open systems – steady and unsteady flow processes.

UNIT II SECOND LAW AND AVAILABILITY ANALYSIS

9+3

Heat Engine – Refrigerator - Heat pump. Statements of second law and their equivalence & corollaries. Carnot cycle - Reversed Carnot cycle - Performance - Clausius inequality. Concept of entropy - T-s diagram - Tds Equations - Entropy change for a pure substance. Ideal gases undergoing different processes - principle of increase in entropy. Applications of II Law. High- and low-grade energy. Availability and Irreversibility for open and closed system processes - I and II law Efficiency.

UNIT III PROPERTIES OF PURE SUBSTANCE

9+3

Steam - formation and its thermodynamic properties - P-v, P-T, T-v, T-s, h-s diagrams. PVT surface. Determination of dryness fraction. Calculation of work done and heat transfer in non-flow and flow processes using Steam Table and Mollier Chart. Application of I and II law for pure substances.

UNIT IV IDEAL AND REAL GASES, THERMODYNAMIC RELATIONS

9+3

Properties of Ideal gas - Ideal and real gas comparison - Equations of state for ideal and real gases - Reduced properties - Compressibility factor - Principle of Corresponding states. Generalised Compressibility Chart and its use. Maxwell relations, Tds Equations, Difference and ratio of heat capacities, Energy equation, Joule-Thomson Coefficient, Clausius Clapeyron equation, Phase Change Processes. Simple Calculations.

UNIT V GAS MIXTURES AND PSYCHROMETRY**9+3**

Mole and Mass fraction, Dalton's and Amagat's Law. Properties of gas mixture – Molar mass, gas constant, density, change in internal energy, enthalpy, entropy and Gibb's function. Psychrometric properties, Psychrometric charts. Property calculations of air vapour mixtures by using chart and expressions. Psychrometric process – adiabatic saturation, sensible heating and cooling, humidification, dehumidification, evaporative cooling and adiabatic mixing. Simple Applications.

TOTAL : 60 PERIODS**COURSE OUTCOMES:**

At the end of the course, students will be able to

- CO1:** Apply the first law of thermodynamics to closed and open systems for thermal equipment.
- CO2:** Analyze the second law of thermodynamics, accurately calculating parameters like work, heat transfer, entropy, and availability for engineering applications.
- CO3:** Apply thermodynamic principles to analyze the properties of pure substance, utilizing diagrams and charts to calculate work, heat transfer, and dryness fraction.
- CO4:** Solve the practical problems by understanding the behavior of Ideal gas and the interrelationship between thermodynamic functions.
- CO5:** Evaluate the properties of gas mixtures and the psychrometric properties for various psychrometric processes.

TEXT BOOKS:

1. Nag P.K., "Engineering Thermodynamics", 6th Edition, Tata McGraw-Hill (2017), New Delhi.

REFERENCES:

1. Cengel Y. and M. Boles, "Thermodynamics - An Engineering Approach", Tata McGraw-Hill, 8th Edition, 2015.
2. Chattopadhyay P., "Engineering Thermodynamics", 2nd Edition, Oxford University Press, 2016.
3. Gordon Rogers, Yon Mayhew, "Engineering Thermodynamics: Work and Heat Transfer", 4th Edition, Pearson, 2002.
4. R.K. Rajput, "A Text Book of Engineering Thermodynamics", 5th Edition, Lakshmi Publications, New Delhi, 2016.

CO's-PO's & PSO's MAPPING

CO	PO												PSO		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	3	2	2	1	-	-	-	-	-	-	-	1	3	-	1
2	3	2	2	1	-	-	-	-	-	-	-	1	3	-	1
3	3	2	2	1	-	-	-	-	-	-	-	1	3	-	1
4	3	2	2	1	-	-	-	-	-	-	-	1	3	-	1
5	3	2	2	1	-	-	-	-	-	-	-	1	3	-	1

1 - low, 2 - medium, 3 - high, '-' - no correlation

PROGRAM SPECIFIC OUTCOMES (PSOs)

The students will be able to

PSO 1: Core Mechanical Proficiency

Analyze, optimize and develop innovative solutions for the real-world mechanical challenges, by applying the principles of design, thermal engineering and computational tools effectively.

PSO 2: Sustainable and Advanced Manufacturing

Apply advanced manufacturing technologies, automation, and sustainable practices to design and execute efficient production systems, ensuring precision, and resource optimization.

PSO 3: Multidisciplinary Integration and Research

Integrate knowledge from various disciplines to solve complex engineering problems, and foster innovation and research in emerging fields such as mechatronics, robotics, Artificial Intelligence (AI) and energy systems.